SOC Home Lab Setup Documentation

Introduction

The purpose of this project is to design and deploy a Security Operations Center (SOC) home lab environment for cybersecurity monitoring, log analysis, and incident detection. The lab environment simulates a small enterprise network using virtual machines hosted on VMware**.**

The environment/VMs includes:

- pfSense firewall/router
- Kali Linux attacker machine
- Windows endpoint system/Victim machine
- Ubuntu Server
- Splunk Enterprise SIEM platform

How I set up the Lab Environment

1. Pfsense

- Downloaded pfsese version 2.8.1 release iso image from the official website https://www.pfsense.org/download/.

Creating pfsense vm

- Using Vmware, I created pfsense vm with the downloaded iso image assigning the RAM: 1 GB, the DISK: 4 GB.

Network Configuration and Installation

- I configured 2 network adapter (pfsense use two adapter, WAN adapter and LAN adapter)
- Assigned IP Address: 192.168.56.1 to the LAN adapter set to Host-Only Network
- WAN (DHCP) set to NAT
- Vlan- not set
- WAN interface set as em0 (default)
- LAN interface set as em1 (default)

After the installation and configuration, i access the pfsense using `https://192.168.56.1`

2. Ubuntu Server

- Downloaded Ubuntu server version 22.04.5 LTS iso image from the official website https://ubuntu.com/download/server
- Using Vmware, I created ubuntu server vm with the downloaded iso image, and assigned the RAM: 4 GB, the DISK: 8 GB.

Configuration and Installation

- I assigned the machine to use DHCP
- Configured openSSH server
- I followed default prompt to set basic settings like Server name,Username, Password

Setting Static IP for Splunk

- I configured netplan and assigned the default gate as 192.168.56.1 (Firewall IP), IP Address: 192.168.56.10 (splunk IP), Interface: ens33

Installation of SPLUNK Enterprise

After downloading splunk enterprise version 10.2.2 from the official website splunk.com/download
I used the following CLI commands to install it in the ubuntu server.

```
sudo dpkg -i splunk-*-linux-10.2.2-amd64.deb
```

After installation I used the following command to start it.

```
sudo /opt/splunk/bin/splunk start --accept-license --run-as-root
```

3. **KALI-ATTACKER**

- Downloaded kali linux version 2025.1 iso image from the official website https://www.kali.org/get-kali/
- Using Vmware, I created kali attacker vm with the downloaded iso image, and assigned the RAM: 2 GB, the DISK: 80 GB.
- configured network adapter 1 as host-only adapter
- After i powered on the kali vm , i assigned static IP Address: 192.168.56.20 to the machine set 192.168.56.1 as default gateway

4. **Windows 10 (Victim Machine)**

- Downloaded windows 10 version 22H2 iso image from the official website https://www.microsoft.com/en-ca/software-download/windows10iso
- Using Vmware, I created Windows 10 vm with the downloaded iso image, and assigned the RAM: 2 GB, the DISK: 100 GB.
- configured network adapter 1 as host-only adapter
- After i powered on the Windows 10 vm , i assigned static IP Address: 192.168.56.30 to the machine set 192.168.56.1 (Pfsense) as default gateway

Network Configuration

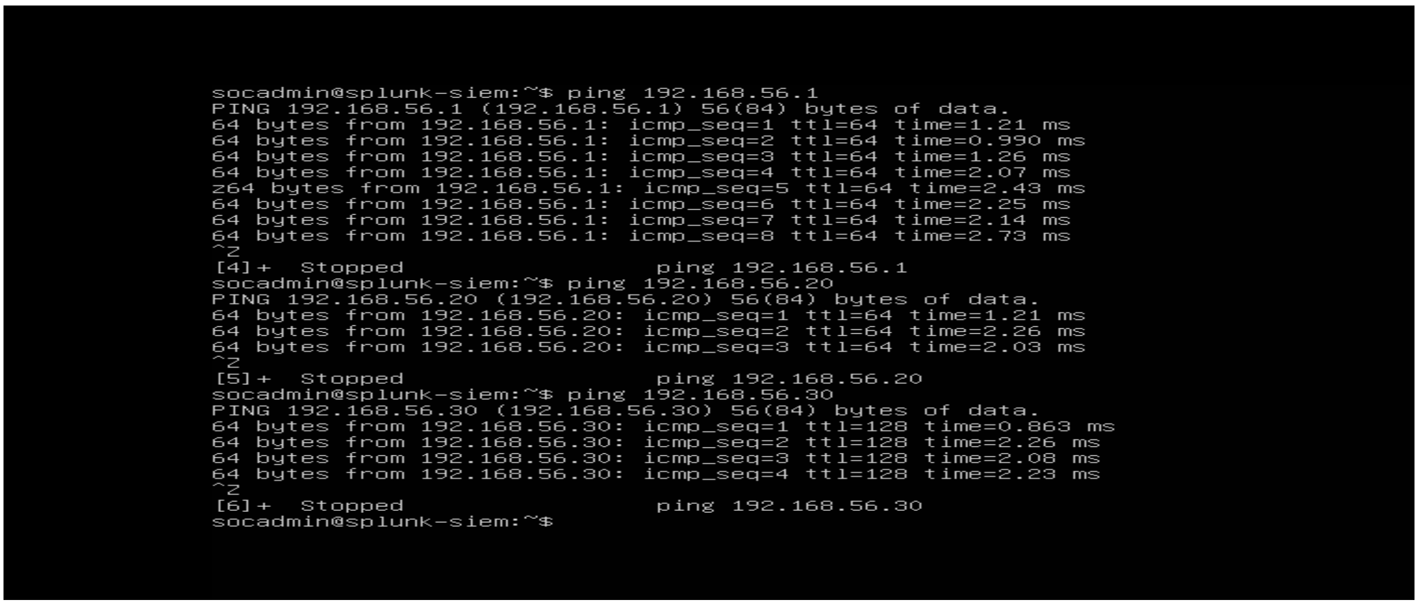
Device	IP Address	Role
Pfsense	192.168.56.1	Firewal/Default Gateway
Splunk	192.168.56.10	SIEM
Kali	192.168.56.20	Attacker Machine
Windows 10	192.168.56.30	Victim Machine

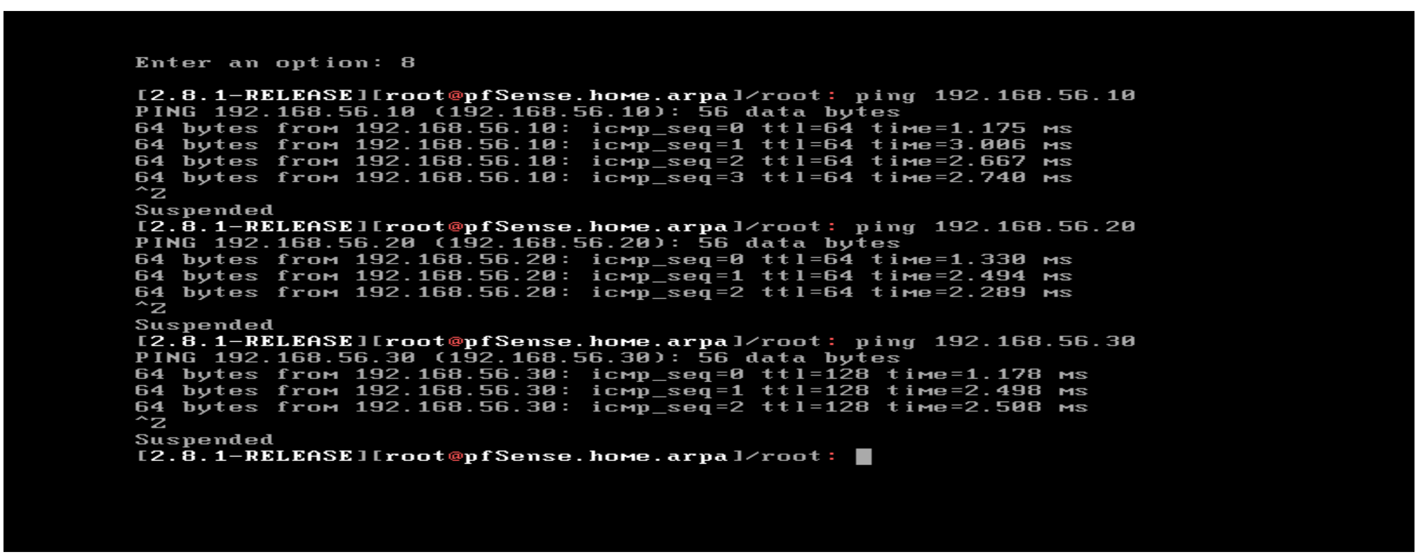
All systems were connected to the same Host-Only VMware network. pfSense acted as the default gateway and firewall for the internal network.

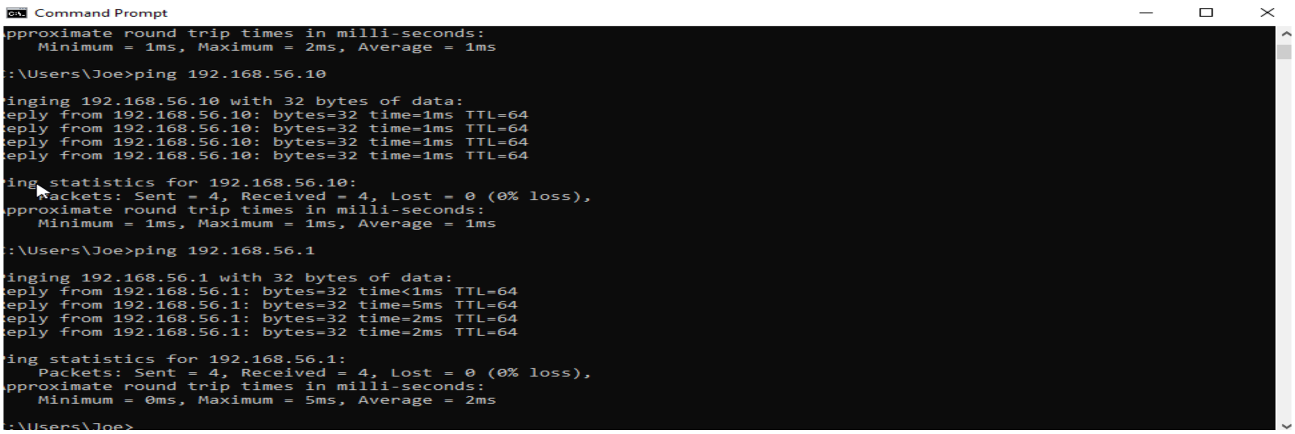
Conclusion

The SOC home lab was successfully deployed using VMware Workstation virtualization. The environment provided practical experience with firewall configuration, SIEM deployment, network monitoring, and log analysis. The lab successfully simulated a small enterprise network environment for cybersecurity training and security operations practice.

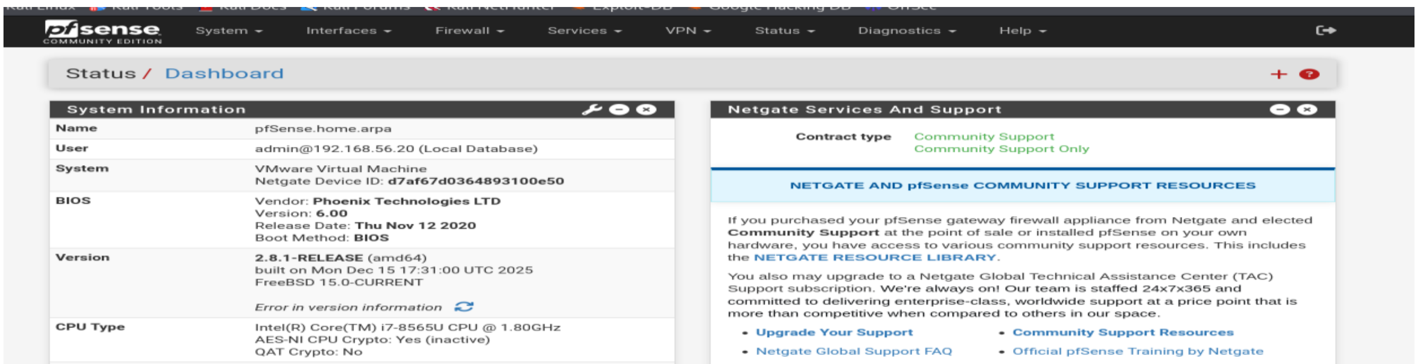
Screenshots of successful Ping



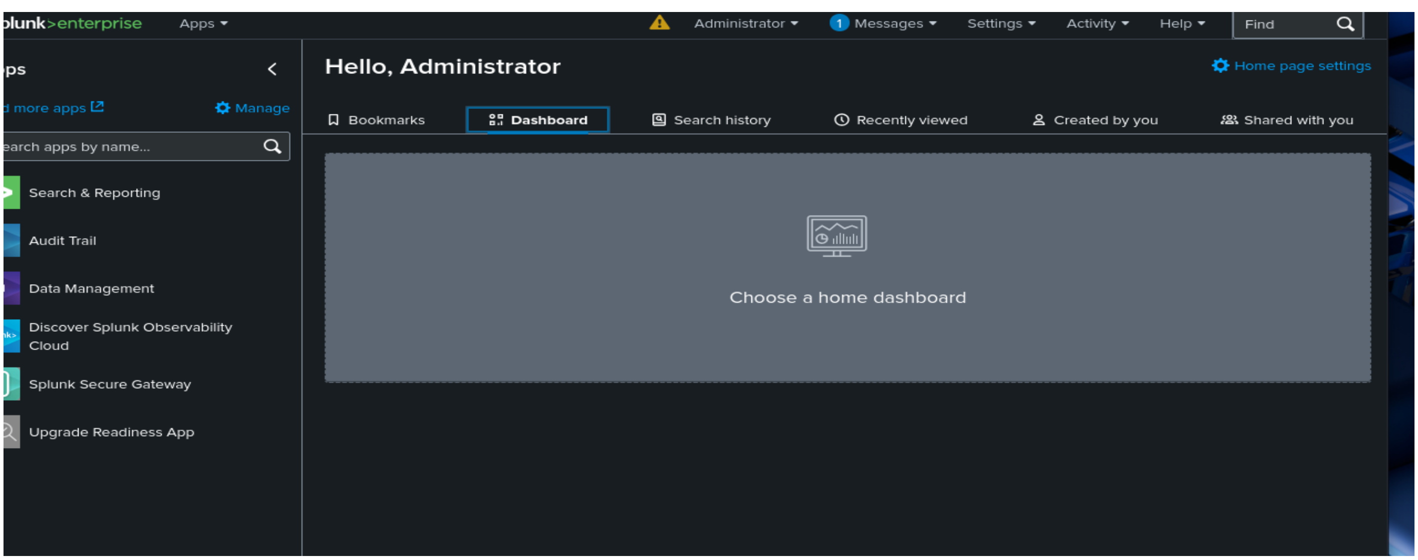


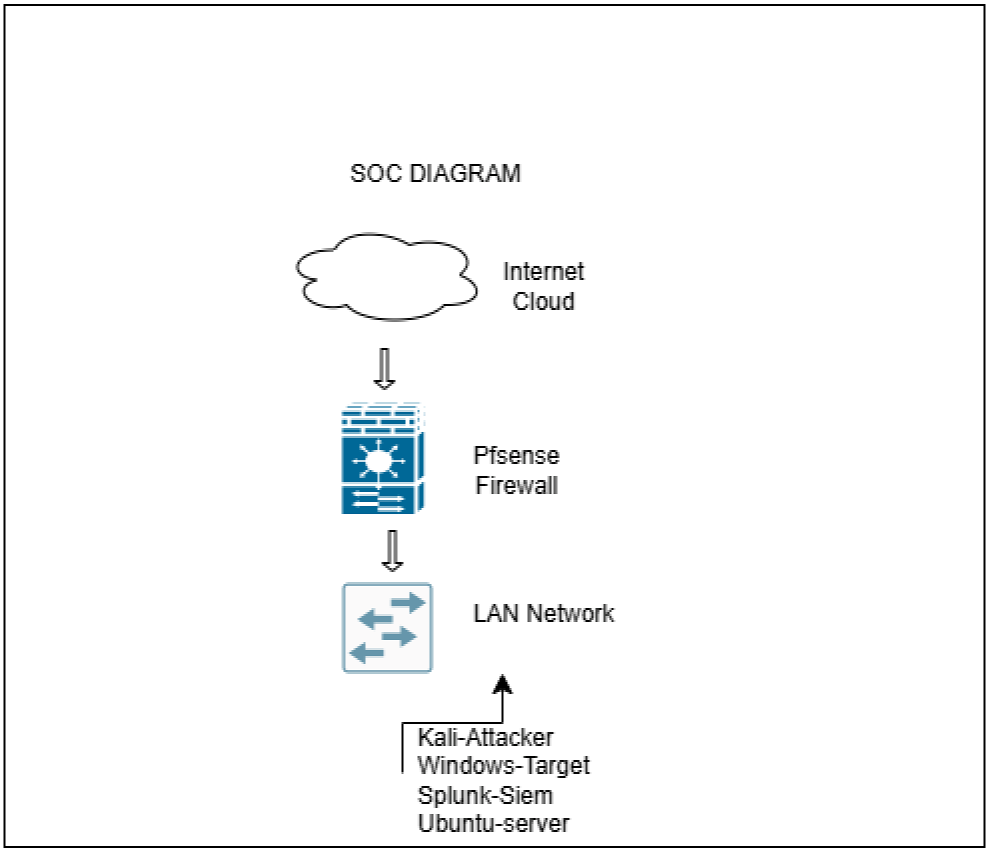


Screenshot of Pfsense Dashboard



Screenshot of Splunk Dashboard





VM DETAILS

Pfsense:
Network:192.168.56.0
IP Address: 192.168.56.1
Version: 2.8.1 RELEASE(amd64)
Specifications: RAM-1G, CPU-Intel Core, Disk-8G

KALI:
Network:192.168.56.0
IP Address: 192.168.56.20
Version: 2025.1
Specifications: RAM-2G, CPU-Intel Core, Disk-80G

WINDOWS 10:
Network:192.168.56.0
IP Address: 192.168.56.30
Version: 22H2 (windows 10 pro)
Specifications: RAM-2G, CPU-Intel Core, Disk-100G

UBUNTU SERVER:
Network:192.168.56.0
IP Address: 192.168.56.10
Version: 22.04.5 LTS
Specifications: RAM-4G, CPU- Intel core i7 , Disk-50G